

Ioannis Mavromatis

LEAD RESEARCH ENGINEER

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🎓 I. Mavromatis Scholar

Highlights

- **ML Systems:** MLOps/LLMOps, Federated & Distributed ML, model monitoring (drift/bias), energy-aware ML pipelines.
- **Cloud-Native Platforms:** Kubernetes (multi-cluster), microservices, GitOps packaging (Helm/kpt), API-driven automation.
- **Distributed Systems (Cloud-Edge Continuum):** orchestration, service placement, resilience/self-healing, scalability.
- **LLMs & Agentic AI:** RAG pipelines, tool-using agents for platform automation, LLM-assisted configuration/YAML generation, telemetry Q&A and self-healing.
- **Observability & Telemetry:** Prometheus/Grafana/Elastic, large-scale metrics/logs/traces, SLO-driven operations.
- **Networking & Systems Security:** secure AI pipelines, intrusion detection & prevention, anomaly detection, networking protocols.
- **Software:** end-to-end life-cycle management, strong development skills.
- **Technical leadership:** platform ownership, architecture decisions, mentoring/line management, stakeholder delivery.

Experience

Digital Catapult

LEAD RESEARCH ENGINEER

London, UK

Nov. 2023 - Present

Projects involved: – REASON / REASON+, SUSTAINET-guarDian, CHEDDAR CHEESE, SONIC, N-CONNECT.

- Leading the **architecture and technical delivery of AI-native orchestration platforms** for cross-domain cloud-edge infrastructures.
- Developing **distributed, cloud-native components** (microservices, APIs, etc.) for intelligent service orchestration and automation.
- Driving the **pre-productisation phase** of our in-house developed intelligent orchestration framework.
- Implementing **agentic AI pipelines** for platform automation, integrating LLM-based planners, tool routing, and structured APIs.
- Building **RAG frameworks over complex relational databases**, enabling natural language querying of platform capabilities and telemetry.
- Engineering **LLM-assisted Kubernetes and YAML generation agents** to automate deployment and QoS-aware configurations.
- Integrating **LLM co-pilots for observability and self-healing** across distributed cloud-native orchestration platforms.
- Designing **schema-aware semantic representations** of platform data to support scalable retrieval and reasoning.
- Developing **energy-efficient ML and Federated Learning pipelines** for resource-constrained cloud-edge environments.
- Performed a **large-scale study of the energy consumption** of Generative and Discriminative AI models.
- Building an **AI Verification, Validation, and Testing (VVT) framework** for ML-enabled distributed and autonomous platforms.
- Architecting **intelligent service placement and migration workflows** across heterogeneous domains (edge, cloud, multi-cluster).
- Driving end-to-end **platform PoCs** from system design to deployment, integration, and large-scale experimentation.
- Assessed the feasibility of **4G/5G nomadic and edge deployments** for connectivity in underserved regions.
- Authoring **bid proposals** across EPSRC, Innovate UK, and Horizon Europe funding bodies.
- Generating **IP and technical publications**.
- **Managing a team of 4 direct reports**, overseeing their career development and technical contributions.

Bristol Research & Innovation Laboratory, Toshiba Europe Ltd.

PRINCIPAL RESEARCH ENGINEER

Bristol, UK

Aug. 2019 - Nov. 2023

*Started as Research Engineer and progressed to Principal level.

Projects involved: – GreenML, UMBRELLA, SYNERGIA, CAVShield, CYTHEMIS.

- Architected and developed a **secure, scalable cloud-native IoT platform** supporting distributed devices and data pipelines.
- Designed **highly-available backend microservices and platform APIs** for large-scale IoT cloud-edge systems.
- Built **data drift detection and monitoring components** for long-running AI/ML systems.
- Researched and implemented **concept drift detection & mitigation strategies** for distributed ML deployments.
- Worked on **IID and non-IID data bias mitigation strategies** for FL training.
- Evaluated and deployed **ML and Federated Learning workloads** using Kubernetes-based MLOps workflows.
- Developed **energy-efficient ML optimisations** (quantisation, power capping, resource-aware training) for MLOps workflows.
- Implemented **federated identity and access management frameworks** for distributed IoT infrastructures.
- Led the development of a **cyber-secure platform** for Cooperative Intelligent Transportation Systems (C-ITS).
- Designed and implemented **distributed IoT testbeds and communication frameworks** for large-scale experimentation.
- Designed and developed an **air quality monitoring solution** for care homes.
- Worked on **secure synchronous flooding solutions** for Bluetooth communications.
- Worked on **end-to-end Intrusion Prevention Systems** evaluations.
- **Led a multidisciplinary team of software engineers** delivering platform-level microservices and system integrations.
- Provided **technical leadership and consultation** to industrial and governmental external partners.
- Led **WP 2 (CAVShield)** including technical strategy, reporting, and presentations to funding bodies and government stakeholders.
- **Mentored** interns, junior engineers, and researchers.
- Generated **IP and technical publications**.

University of Bristol

HONORARY RESEARCH ASSOCIATE

Bristol, UK

Aug. 2019 - Sep. 2023

- Migrated **AI-assisted Intelligent Transportation Solutions** to Cloud-native and Mobile Edge Infrastructures.
- Contributed to research on **distributed AI systems and edge-cloud continuum architectures**.

University of Bristol

RESEARCH ASSOCIATE (Post-Doc)

Bristol, UK

Oct. 2018 - Aug. 2019

Projects involved: – NG-CDI, FLOURISH.

- Developed an open-source **AI-assisted 5G simulator** for C-ITSs (DRIVE framework).
- Developed a **Fog Computing platform** for large-scale connected vehicle service experimentation.
- Implemented and deployed a large-scale **prototyped ITS-G5 testbed**.
- Conducted several **large-scale city-wide vehicular trials** campaigns.
- Designed an **active jammer** - tested performance/disruptions on prototyped ITS-G5 testbed.

University of Bristol

RESEARCH ASSISTANT

Bristol, UK

Feb. 2016 - Aug. 2018

Projects involved: – FLOURISH, VENTURER.

- Developed a **ITS-G5 vehicular communications PoC** platform.
- Conducted **city-wide connected vehicle trials** and analysed the experimental data.
- Developed a **parallelised** (multi-core compatible) **version of the OMNeT++ INET** framework.
- "Virtual Driving Testbed" - developed an **in-lab testbed emulation solution** for vehicular communication testing.

University Carlos III of Madrid (UC3M)

NETWORK RESEARCH ASSISTANT (INTERN)

Madrid, Spain

Apr. 2012 – Sep. 2012

Project involved: FLAVIA.

- Implemented **energy-efficiency optimisations** for **WLAN networks** (IEEE 802.11b/g/n/u).
- Modified **IEEE 802.11b/g drivers and firmware** (OpenWRT) to improve power consumption.
- Prototyped and evaluated a **custom wireless testbed** to assess system performance and energy impact.

Education

Dept. of Electrical & Electronic Engineering, University of Bristol

PHD IN COMMUNICATIONS

Bristol, UK

Oct. 2014 - Oct. 2018

- **PhD Title:** "5G Communication Framework for Smarter Autonomous Vehicles".
- **1st Year's Research Group Project:** "Localisation Of Strayed Targets (LOST)" (taught phase).

Dept. of Electrical & Electronic Engineering, University of Bristol

MSc IN "WIRELESS COMMUNICATIONS AND SIGNAL PROCESSING"

Bristol, UK

Sep. 2013 - Sep. 2014

- **MSc Thesis:** "Propagation Modelling using Iterative Physical Optics".

Dept. of Information Technology, ATEI of Thessaloniki

BSc IN "INFORMATION TECHNOLOGY ENGINEERING"

Thessaloniki, Greece

May. 2007 - Nov. 2011

- **BSc Thesis:** "Studying Quality of Experience (QoE) over Wireless Networks".

Skills

LLMs / RAG	LangChain, Transformers, vLLM, Ollama, exo, prompt/tool routing, vector search/embeddings, RAG evaluation.
ML/FL	PyTorch, NumPy, Flower FL, MLflow, KubeFlow.
Programming	Python (platform/ML/automation), Go (services), C/C++ (performance-critical systems).
Cloud/Edge	Kubernetes (K8s, K3s, MicroK8s), Helm/kpt, Docker, multi-cluster patterns, API gateways (Nginx/HAProxy).
Data	PostgreSQL, InfluxDB, Redis, MySQL; data pipelines for telemetry/analytics.
Observability	Prometheus, Grafana, Elastic, Kibana, Fluentd/FluentBit.
Automation/CI	Git, GitHub Actions/Jenkins, Infrastructure automation (TravisCI, Ansible).
Virtualisation	Proxmox, OpenStack, Google Cloud Platform (GCP), Amazon Web Services (AWS).
Networking	OAuth2, OpenVPN, Wireshark/Tcpdump, Nmap, FTP, LDAP.
Simulators	OMNeT++ (including Veins), SUMO, NS-2, NS-3, OPNET.
OSs	Linux-based Distributions (including kernel and driver programming), OS X, MS Windows.

Patents & Selected Publications

05/2025	A. Sanchez-Mompo, I. Mavromatis , P. Li, K. Katsaros, and A. Khan, "Green MLOps to Green GenOps: An Empirical Study of Energy Consumption in Discriminative and Generative AI Operations"	Information MDPI
12/2024	I. Mavromatis , S. De Feo, and A. Khan, "FLAME: Adaptive and Reactive Concept Drift Mitigation for Federated Learning Deployments"	EWSN 2024
07/2024	I. Mavromatis , K. Katsaros, and A. Khan, "Computing Within Limits: An Empirical Study of Energy Consumption in ML Training and Inference"	ICEST 2024
03/2024	J. Gao, I. Mavromatis , P. Li, P. Carnelli, and A. Khan, "Mitigating System Bias in Resource Constrained Asynchronous Federated Learning Systems"	IEEE PerCom 2024
11/2023	O. Belarbi, T. Spyridopoulos, E. Anthi, I. Mavromatis , P. Carnelli and A. Khan, "Federated Deep Learning for Intrusion Detection in IoT Networks"	IEEE GLOBECOM 2023
11/2023	I. Mavromatis , S. De Feo, P. Carnelli, R. J. Piechocki, and A. Khan, "FROST: Towards Energy-efficient AI-on-5G Platforms - A GPU Power Capping Evaluation"	IEEE CSCN 2023
11/2023	H. Friji, I. Mavromatis , A. Sanchez-Mompo, P. Carnelli, A. Olivereau, and A. Khan, "Multi-stage Attack Detection and Prediction Using Graph Neural Networks: An IoT Feasibility Study"	IEEE TrustCom 2023
06/2023	T. Chow, U. Raza, I. Mavromatis , and A. Khan, "FLARE: Detection and Mitigation of Concept Drift for Federated Learning based IoT Deployments"	IEEE IWCMC 2023
01/2023	I. Mavromatis , A. Sanchez-Mompo, F. Raimondo, J. Pope, M. Bullo, I. Weeks, V. Kumar, P. Carnelli, G. Oikonomou, T. Spyridopoulos, and A. Khan, "LE3D: A Lightweight Ensemble Framework of Data Drift Detectors for Resource-Constrained Devices"	IEEE CCNC 2023
10/2022	C. Lockie, I. Mavromatis , A. Stanoev, Y. Jin, and G. Oikonomou, "Securing Synchronous Flooding Communications: An Atomic-SDN Implementation"	EWSN 2022
04/2022	I. Mavromatis , A. Stanoev, A. J. Portelli, C. Lockie, M. Ammann, Y. Jin, and M. Sooriyabandara, "Reliable IoT Firmware Updates: A Large-scale Mesh Network Performance Investigation"	IEEE WCNC 2022
03/2022	US449398, "System and Method for Detecting and Rectifying Concept Drift in Federated Learning"	U.S. Patent
09/2019	X. Wang, I. Mavromatis , A. Tassi, R. Santos-Rodriguez, and R. J. Piechocki, "Location Anomalies Detection for Connected and Autonomous Vehicles"	IEEE CAVS 2019
04/2019	A. Tassi, I. Mavromatis , R. J. Piechocki, and A. Nix, "Secure Data Offloading Strategy for Connected and Autonomous Vehicles"	IEEE VTC 2019

Honors & Awards

09/2024	AI Award for High Tech & Telecom, for REASON project , National AI Awards, Software/Network Architect
09/2022	Industrial Innovation Award, for UMBRELLA project , Connected Britain, Lead Backend Architect / Developer
05/2019	IEEE Best Paper Award , IEEE Vehicular Technology Conference (IEEE VTC-Spring)
12/2018	IEEE Popularity Award , IEEE Vehicular Networking Conference (IEEE VNC)

Training & Certificates

08/2020	Kubernetes MasterClass : Kubernetes & Docker Swarm for DevOp , Udemy
08/2020	Machine Learning , Coursera, organised by Stanford University

Volunteering

- Technical Program Committee member and reviewer - several conferences and journals (2017 - now).
- Industrial Mentoring Scheme, Faculty of Engineering, University of Bristol (2019 - 2022).